

Assignment No	2 & 3
Assignment Title	Paper Review
Mini Project Title	RAGTrust AI: Explainable Claim-Level Hallucination Detection and Evidence-Grounded Verification
Team Members Name	Rishi Phadale Madhura Vilas Suroshe Asmee Jadhav Krupa Gurav
Roll No's	23102C0070 24102C2002 24102C2003 24102C2005
Paper1 Title	RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models
Authors	Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, Tong Zhang
Journal link and Year of Publication	ACL 2024, pp. 10862-10878 - https://aclanthology.org/2024.acl-long.585/
Description	RAGTruth is the closest dataset-level foundation for this project because it studies hallucinations specifically inside retrieval-augmented generation. The corpus contains nearly 18,000 naturally generated RAG responses from multiple LLMs and includes careful hallucination annotations at both response and word levels. The paper shows that retrieval does not eliminate hallucination: a model can still produce unsupported or contradictory statements even when relevant evidence has been retrieved. For RAGTrust AI, this motivates two design choices: using RAGTruth as the primary supervised benchmark for the response-level detector, and moving beyond a single answer-level label toward localization of the exact unsupported span. Our current CatBoost detector is trained from engineered semantic, NLI, lexical/entity and linguistic signals derived from this corpus, while the enhanced application extends the benchmark setting into document-grounded claim auditing.
Paper2 Title	VeriScore: Evaluating the Factuality of Verifiable Claims in Long-Form Text Generation
Authors	Yixiao Song, Yekyung Kim, Mohit Iyer
Journal link and Year of Publication	Findings of EMNLP 2024, pp. 9447-9474 - https://aclanthology.org/2024.findings-emnlp.552/
Description	VeriScore addresses a limitation that becomes important when evaluating long answers: not every sentence is equally factual or even independently verifiable. Instead of scoring a long generation as one block, the method extracts verifiable claims and checks them against external evidence. Human evaluation in the paper indicates that its claim extraction is more sensible across diverse long-form tasks than earlier factuality metrics. This directly supports RAGTrust AI's document-audit mode, where a paragraph is decomposed into atomic claims before verification. The key difference is that RAGTrust uses a trusted user-supplied document as the evidence space and combines retrieval, DeBERTa NLI, factual mismatch checks and a learned CatBoost signal, rather than relying only on a factuality metric. VeriScore therefore motivates claim-level granularity and explicit handling of claims that cannot be verified from the available evidence.
Paper3 Title	Optimizing Decomposition for Optimal Claim Verification
Authors	Yining Lu, Noah Ziems, Hy Dang, Meng Jiang

Journal link and Year of Publication	ACL 2025, pp. 5095-5114 - https://aclanthology.org/2025.acl-long.254/
Description	This paper studies the Decompose-Then-Verify paradigm and shows that claim decomposition and claim verification cannot be treated as independent modules. The authors introduce atomicity as a measure of information density and show that a poorly decomposed statement can reduce downstream verification accuracy. Their dynamic decomposition framework learns a decomposition policy using verifier feedback and reports improvements in verification confidence and accuracy across different verifiers and datasets. This work is especially relevant to RAGTrust AI because the application must audit long paragraphs, where a single sentence may contain several factual relations, dates or numbers. It motivates evaluating alternative claim-splitting policies rather than assuming sentence boundaries are sufficient. In the proposed research extension, claim atomicity becomes an explicit experiment: sentence-level splitting, rule-based atomic splitting and verifier-aware decomposition can be compared using claim-level F1 and contradiction recall.
Paper4 Title	FaStFact: Faster, Stronger Long-Form Factuality Evaluations in LLMs
Authors	Yingjia Wan, Haochen Tan, Xiao Zhu, Xinyu Zhou, Zhiwei Li, Qingsong Lv, Changxuan Sun, Jiaqi Zeng, Yi Xu, Jianqiao Lu, Yinhong Liu, Zhijiang Guo
Journal link and Year of Publication	Findings of EMNLP 2025 - https://aclanthology.org/2025.findings-emnlp.1295/
Description	FaStFact focuses on two practical weaknesses of long-form factuality pipelines: inefficient claim-by-claim processing and weak evidence collection. It keeps the decompose-then-verify pattern but adds chunk-level claim extraction, confidence-based pre-verification and document-level evidence retrieval so that fewer expensive searches and verifier calls are needed. This is highly relevant to RAGTrust AI's target use case of auditing long PDFs and long AI responses. The project follows the same efficiency principle by chunking large documents once, embedding them with MiniLM, retrieving only the top-k evidence chunks for each claim, and running NLI on the small retrieved evidence set instead of the complete document. The research opportunity is to measure the trade-off between retrieval depth, verification accuracy and latency, which is important for a hackathon-ready system as well as for a deployable RAG trust layer.
Paper5 Title	HaluEval: A Large-Scale Hallucination Evaluation Benchmark for Large Language Models
Authors	Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, Ji-Rong Wen
Journal link and Year of Publication	EMNLP 2023, pp. 6449-6464 - https://aclanthology.org/2023.emnlp-main.397/
Description	HaluEval provides a large benchmark for studying whether language models can recognize hallucinated content. It contains 35,000 hallucinated/normal examples, including task-specific samples for question answering, knowledge-grounded dialogue and summarization, and the authors show that hallucination recognition remains difficult while external knowledge and reasoning can improve it. In RAGTrust AI, HaluEval is used as an external zero-shot generalization benchmark rather than a retraining source. A RAGTruth-trained CatBoost detector is evaluated on a deterministic pair-preserving HaluEval subset with no fine-tuning or threshold tuning, allowing the project to study how well learned hallucination signals transfer to a new benchmark and to an unseen dialogue task. This paper therefore anchors the project's cross-dataset validation and motivates the broader claim that a practical detector should be tested outside its training distribution.
Paper6 Title	FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation
Authors	Sewon Min, Kalpesh Krishna, Xinxu Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, Hannaneh Hajishirzi
Journal link and Year of Publication	EMNLP 2023, pp. 12076-12100 - https://aclanthology.org/2023.emnlp-main.741/

Description	FactScore is the paper that originated the atomic-fact decomposition approach that VeriScore, FaStFact and the Decompose-Then-Verify literature all build on. It defines an atomic fact as the smallest independently checkable unit of a long-form generation and scores factual precision as the fraction of atomic facts supported by a chosen knowledge source, rather than assigning one label to an entire response. It also shows that a small automated verifier can approximate costly human factual-precision judgments with well under 2% error, which is the same trade-off RAGTrust AI relies on when it substitutes a retrieval and NLI pipeline for exhaustive human fact-checking. For RAGTrust AI, FactScore motivates treating the trusted document (rather than Wikipedia) as the knowledge source against which atomic claims are checked, and it anchors the project's precision-style metric for how many extracted claims are actually supported by retrieved evidence.
Paper7 Title	MiniCheck: Efficient Fact-Checking of LLMs on Grounding Documents
Authors	Liyan Tang, Philippe Laban, Greg Durrett
Journal link and Year of Publication	EMNLP 2024, pp. 8818-8847 - https://aclanthology.org/2024.emnlp-main.499/
Description	MiniCheck tackles the same core sub-problem RAGTrust AI relies on for claim verification: deciding whether a piece of generated text is grounded in a supplied document. Instead of repeatedly prompting a large LLM to check each claim, the authors train small entailment-style models on synthetic, structurally realistic factual errors and show these models match GPT-4-level accuracy at roughly 400x lower inference cost, evaluated on a unified benchmark, LLM-AggreFact, that includes RAGTruth. This is directly relevant to RAGTrust AI's design choice of using a lightweight DeBERTa NLI model rather than repeated LLM calls to check each claim against retrieved evidence chunks, and it supports the project's efficiency argument that grounding checks for long documents must stay cheap enough for interactive auditing. MiniCheck also motivates a future ablation comparing the project's current NLI-plus-CatBoost verifier against a MiniCheck-style trained checker on the same evidence chunks.
Summary of all papers	The seven reviewed works collectively define the research space that RAGTrust AI targets. RAGTruth demonstrates that hallucinations remain a problem even in retrieval-augmented systems and provides response- and word-level annotations suitable for supervised learning. HaluEval supplies an independent benchmark for cross-dataset evaluation. FactScore originated the atomic-fact decomposition and knowledge-source-relative scoring that the rest of this literature builds on, and shows that a lightweight automated verifier can substitute for costly human fact-checking with low error. VeriScore extends this line of work by decomposing long-form generations into verifiable claims instead of assigning one factuality score to an entire paragraph. Lu et al. (2025) further show that claim atomicity directly affects downstream verification quality, while FaStFact demonstrates that efficient long-form verification requires selective claim processing and strong document-level evidence retrieval. MiniCheck complements this by showing that small, purpose-trained entailment models can match large-LLM-level grounding accuracy at a fraction of the inference cost, which supports using lightweight NLI models for claim verification instead of repeated LLM calls. Research gap: existing work usually optimizes one layer of the problem - benchmark construction, factuality scoring, claim decomposition, or evidence retrieval. A user-facing RAG system, however, needs these capabilities together. RAGTrust AI therefore combines (1) a learned response-level hallucination detector, (2) claim-level decomposition, (3) retrieval over large trusted documents, (4) NLI-based support/contradiction verification, (5) entity/number/date consistency checks, (6) evidence localization and correction, and (7) explainability and drift analysis. The research focus is not merely to output a hallucination probability, but to determine whether a hybrid response-level + claim-level framework improves localization, contradiction recall and cross-domain robustness while remaining efficient enough for interactive document auditing.
Problem Statement for Mini Project	Large Language Models and Retrieval-Augmented Generation systems can produce long, fluent answers that mix correct statements with a small number of unsupported or contradictory claims. A single response-level hallucination probability is insufficient for real users because it does not identify which claim

	is wrong, what source evidence contradicts it, or whether the error involves an entity, number, date or factual relation. The problem becomes harder for large PDFs because the verifier cannot compare every generated claim against the full document directly. RAGTrust AI proposes an explainable claim-level verification framework for long-form LLM/RAG outputs. The system chunks and indexes trusted documents with MiniLM embeddings, decomposes an AI answer into atomic claims, retrieves the most relevant evidence for every claim, and combines DeBERTa NLI, semantic/lexical signals, entity-number-date consistency checks and a RAGTruth-trained CatBoost detector. Each claim is classified as Supported, Unsupported or Contradicted, suspicious text is highlighted, relevant source evidence is shown, and a grounded correction can be proposed. The research evaluates supervised performance on RAGTruth, zero-shot generalization on HaluEval, feature importance/ablation, statistical confidence, distribution drift, and a planned claim-level benchmark for long-document hallucination localization.
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